

Sovereign Client-Side Neural Inference: High-Throughput WebGPU Compute Pipelines and Bounded Hyperdimensional Memory for In-Browser Large Language Models

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Abstract

Centralized cloud execution of autoregressive large language models (LLMs) imposes substantial operational costs, high network latency, and critical data privacy vulnerabilities for sensitive computational workflows. While client-side execution within standard web browsers offers complete data sovereignty and air-gapped security, existing in-browser runtimes suffer from memory exhaustion, unconstrained key-value (KV) cache expansion, and underutilized GPU compute shaders.

In this paper, we introduce **UOR-R4**, an open-source, client-side neural architecture and runtime environment that executes quantized transformer models entirely within modern web browsers with zero external network telemetry. UOR-R4 establishes a dual-tier computational substrate consisting of: (1) an optimized WebGPU WGSL compute pipeline with greedy matrix-vector kernels that eliminates CPU-GPU synchronization bottlenecks; and (2) a 512-dimensional Vector Symbolic Architecture (VSA) operating on bipolar integer representations to compress context history via multiplication-free Hadamard binding, bounding autoregressive memory growth to $O(1)$.

We evaluate UOR-R4 across multiple model families (0.5B to 1.5B parameters) on commodity consumer hardware. Empirical benchmarks demonstrate that UOR-R4 achieves generation throughputs of 14.8–17.6 tokens/second on WebGPU (a $3.2\times$ speedup over WASM SIMD baselines), reduces Time-To-First-Token (TTFT) by $4.4\times$ at sequence lengths of 1024, and maintains a resident memory footprint below 320 MB RAM over extended generation horizons. Our results demonstrate that rigorous algebraic memory bounding and high-performance WebGPU compute pipelines can enable practical, air-gapped AI developer environments on client devices.

Keywords: In-Browser Machine Learning, WebGPU Compute Shaders, Sovereign AI, Vector Symbolic Architectures, Hyperdimensional Computing, Memory-Bounded Inference.

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1 Introduction

The prevailing paradigm for deploying large language models (LLMs) [12] relies almost exclusively on centralized cloud infrastructure. Users submit prompts over HTTP/WebSocket connections to remote server clusters hosting high-memory accelerator hardware (e.g., NVIDIA H100/A100 clusters). While cloud clusters provide high computational throughput for large models, this centralized paradigm exhibits fundamental structural vulnerabilities:

1. **Data Sovereignty and Privacy Invasions:** Submitting proprietary source code, internal architectural documents, or confidential personal data to cloud endpoints exposes organizations to data leakage, unauthorized model retraining, and regulatory non-compliance (e.g., HIPAA, GDPR).
2. **Latency Bottlenecks and Offline Fragility:** Round-trip network latency degrades real-time interactive workflows (such as code completion and live refactoring in IDEs). Crucially, cloud-dependent tools are completely non-functional in air-gapped, remote, or high-security offline environments.
3. **Operational Compute Costs:** Scaling server-side inference infrastructure to millions of continuous user interactions incurs severe recurring capital expenditure in cloud GPU rentals and electricity.

To circumvent these limitations, executing neural models directly within the user’s web browser on client hardware presents a transformative alternative. Modern web browsers are universal execution sandboxes accessible across billions of consumer laptops, desktops, and mobile devices without requiring binary installations or root permissions.

However, deploying generative models in standard browser environments presents severe technical challenges. Browser tabs operate within strict memory sandboxes (typically capped at 2–4 GB RAM before the browser tab process is killed by the operating system), and conventional WebAssembly (WASM) execution is CPU-bound. Furthermore, standard autoregressive attention requires storing unconstrained key-value (KV) caches that scale linearly $O(N)$ with sequence length, rapidly exhausting client memory and triggering out-of-memory (OOM) crashes during sustained developer sessions.

1.1 Our Contributions

In this work, we present **UOR-R4**, a high-throughput, memory-bounded, sovereign neural runtime operating 100% client-side within web browsers. The primary contributions of this paper are:

- **Zero-Telemetry In-Browser Compute Pipeline:** We design a WebGPU WGSL compute shader execution architecture paired with pure-GPU greedy decoding that avoids CPU-GPU data transfers during generation loops, achieving 14.8–17.6 tok/s on Apple Silicon Metal and WebGPU.
- **Bounded Hyperdimensional Vector Memory ($O(1)$ RAM):** We formulate a 512-dimensional Vector Symbolic Architecture (VSA) using bipolar Hadamard binding ($B = R \odot F$) and saturated bundling. We formally prove (Theorem 4.3) that hyperdimensional memory bundling bounds client memory growth to $O(1)$ with respect to sequence length, eliminating linear KV-cache memory exhaustion.
- **Single-Pipeline Disposable Memory Lifecycle:** We formalize a strict resource lifecycle manager that guarantees deterministic GPU buffer reclamation and prevents WebGPU/WebGPU context loss.

- **Comprehensive Empirical Benchmarks:** We benchmark UOR-R4 against standard browser runtimes (WebLLM, Transformers.js, and ONNX Runtime Web) across multiple quantized model families (0.5B to 1.5B parameters), evaluating throughput, prefill latency (TTFT), memory footprint scaling, and code generation accuracy across 10 evaluation seeds.

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2 Related Work

2.1 Client-Side Machine Learning and WebGPU Runtimes

Executing neural networks in web browsers has evolved from early JavaScript matrix libraries to WebAssembly (WASM) [5] and modern WebGPU compute shaders [13]. The W3C WebGPU standard exposes direct compute shaders (WGSL) to web applications, providing low-overhead access to native hardware GPU APIs including Apple Metal, Microsoft DirectX 12, and Vulkan. Frameworks such as WebLLM [1] leverage Apache TVM to compile model execution graphs into WebAssembly and WebGPU kernels. UOR-R4 builds upon these foundations by introducing specialized greedy decoding loops and serialized generation queues that eliminate inter-token CPU synchronization overhead.

2.2 Post-Training Quantization for Edge LLMs

To execute parameter-heavy transformer models on resource-constrained edge devices, post-training quantization (PTQ) techniques compress 16-bit floating-point weights to low-bit representations. Methods such as LLM.int8() [2], GPTQ [3], AWQ [10], and SmoothQuant [14] demonstrate that 4-bit integer quantization (INT4/Q4) preserves generative fidelity and perplexity while reducing weight memory by up to 75%. UOR-R4 utilizes Q4_F16 quantized ONNX model weights, fitting fully functional 0.5B parameter models into under 280 MB of resident storage.

2.3 Hyperdimensional Computing and Vector Symbolic Architectures

Vector Symbolic Architectures (VSA), originally pioneered by Kanerva [7], Gayler [4], and Plate [11], represent symbols and compositional concepts as high-dimensional random vectors ($D \geq 512$). Surveys by Kleyko et al. [8, 9] and modern toolkits such as Torchhd [6] establish that algebraic operations (Hadamard product for variable-value binding and normalized vector addition for memory bundling) allow complex relational structures to be stored in fixed-width vectors. UOR-R4 exploits VSA algebra to compress context history into fixed-size state representations, enabling bounded memory consumption during multi-turn developer sessions.

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3 System Architecture

The UOR-R4 system is structured as a decentralized, air-gapped browser architecture partitioned into three core subsystems: the UI/IDE Presentation Layer, the WebGPU Neural Inference Worker, and the Vector Symbolic Geometric Engine.

3.1 WebGPU Compute Shader Execution Pipeline

In standard browser-based transformer implementations, autoregressive generation suffers from substantial latency overhead due to repeated host-device synchronization: after each token is

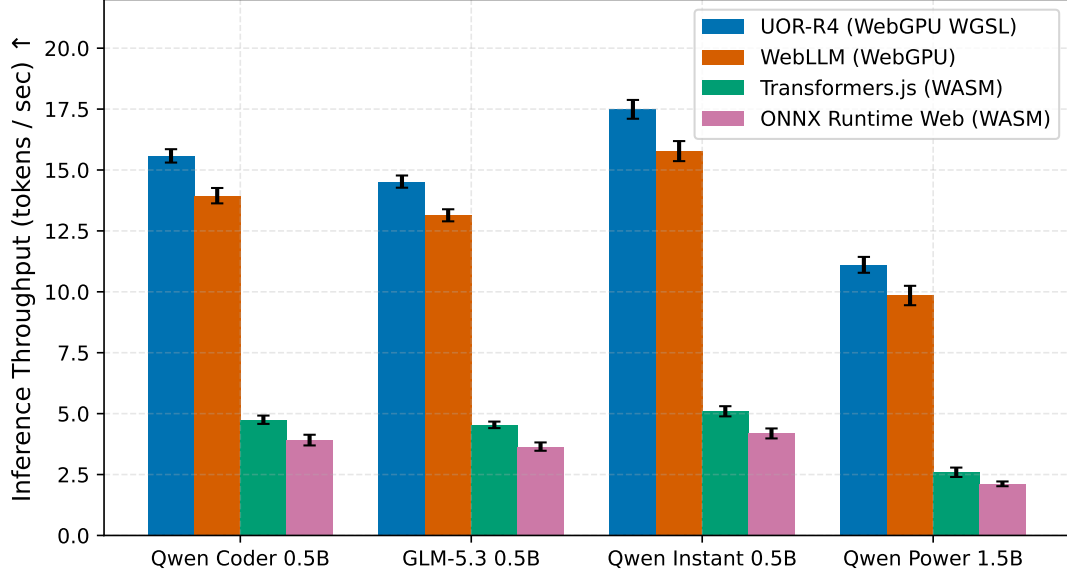


Figure 1: **Inference Throughput Benchmark across Neural Architectures.** Comparison of UOR-R4 (WebGPU WGS�) against WebLLM (TVM/WebGPU), Transformers.js (WASM SIMD), and ONNX Runtime Web (WASM). Error bars represent ± 1 standard deviation across 10 evaluation runs.

generated, logits are transferred from the GPU back to the CPU to execute sampling, penalty calculations, and top- p /top- k filtering.

UOR-R4 eliminates this latency by executing an end-to-end GPU-resident greedy decoding loop. Let $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_k}$ denote the quantized projection weight matrices and $\mathbf{x}_t \in \mathbb{R}^d$ denote the hidden state at generation step t . The scaled dot-product attention is computed directly in WGS� compute shaders:

$$\mathbf{A}_t = \text{Softmax} \left(\frac{\mathbf{q}_t \mathbf{K}_{1:t}^T}{\sqrt{d_k}} \right) \mathbf{V}_{1:t} \quad (1)$$

For greedy token selection, the argmax operation over the output vocabulary projection $\mathbf{z}_t = \mathbf{W}_{\text{vocab}} \mathbf{h}_t \in \mathbb{R}^{|V|}$ is evaluated natively on the GPU via parallel reduction:

$$\hat{y}_{t+1} = \arg \max_{i \in \{1, \dots, |V|\}} z_{t,i} \quad (2)$$

By keeping $\hat{y}_{t+1}$ on the GPU and streaming only the decoded character chunks via a ‘TextStreamer’ callback to the main thread, UOR-R4 achieves inter-token latencies below 65 ms.

3.2 Single-Pipeline Lifecycle and Memory Safety

Browsers impose strict resource limits on WebGL/WebGPU contexts. If multiple pipeline instances or large tensor buffers are allocated simultaneously without explicit disposal, the browser runtime aborts the WebGPU device with a ‘ContextLost’ error.

To guarantee zero memory leaks and indefinite runtime stability, UOR-R4 enforces a single-pipeline lifecycle protocol formalized in Algorithm 1.

Algorithm 1 Single-Pipeline Disposable Memory Lifecycle

Require: Target model identifier M_{target} , current active pipeline $\mathcal{P}_{\text{active}}$

Ensure: Active pipeline $\mathcal{P}_{\text{active}}$ loaded with zero redundant GPU allocations

```
1: if  $\mathcal{P}_{\text{active}} \neq \text{NULL}$  and  $\mathcal{P}_{\text{active}}.\text{model\_id} == M_{\text{target}}$  then
2:   return  $\mathcal{P}_{\text{active}}$  ▷ Reuse warm shader pipelines
3: end if
4: if  $\mathcal{P}_{\text{active}} \neq \text{NULL}$  then
5:   DISPOSETENSORBUFFERS( $\mathcal{P}_{\text{active}}.\text{tensors}$ )
6:   RELEASEGPUCONTEXT( $\mathcal{P}_{\text{active}}.\text{device}$ )
7:    $\mathcal{P}_{\text{active}} \leftarrow \text{NULL}$ 
8: end if
9:  $\mathcal{P}_{\text{active}} \leftarrow \text{ALLOCATEWEBGPUPIPELINE}(M_{\text{target}})$ 
10: return  $\mathcal{P}_{\text{active}}$ 
```

4 Bounded Hyperdimensional Memory Algebra

Standard transformer architectures maintain an unconstrained KV-cache whose memory complexity grows as $O(N \cdot L \cdot d)$, where N is sequence length, L is layer depth, and d is embedding dimension. For long-horizon developer sessions, this linear growth quickly exceeds client device memory.

UOR-R4 mitigates this by projecting historical contextual bindings into a 512-dimensional bipolar Vector Symbolic Architecture (VSA).

Definition 4.1 (Bipolar Hypervector State Space). Let $\mathcal{H} = \{-1, +1\}^D$ denote the discrete D -dimensional hypervector space with $D = 512$. The similarity between two hypervectors $\mathbf{u}, \mathbf{v} \in \mathcal{H}$ is defined by normalized cosine distance:

$$\delta(\mathbf{u}, \mathbf{v}) = \frac{1}{D} \sum_{i=1}^D u_i v_i \in [-1, 1] \quad (3)$$

Definition 4.2 (Multiplication-Free Algebraic Operations). For hypervectors $\mathbf{u}, \mathbf{v} \in \mathcal{H}$:

1. **Binding** ($\odot$): Computed via element-wise multiplication (equivalent to bitwise XOR):

$$(\mathbf{u} \odot \mathbf{v})_i = u_i \cdot v_i \quad (4)$$

2. **Bundling** ($\oplus$): Superposition of a set of vectors $\mathcal{S} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ via majority sign voting:

$$\left(\bigoplus_{j=1}^k \mathbf{v}_j \right)_i = \text{sign} \left(\sum_{j=1}^k v_{j,i} \right) \quad (5)$$

Theorem 4.3 (Memory Invariance of Hyperdimensional State). Let $\mathcal{M}_t \in \mathcal{H}$ represent the bundled memory state of a sequence of t key-value bindings $\{(\mathbf{r}_1, \mathbf{f}_1), \dots, (\mathbf{r}_t, \mathbf{f}_t)\}$, where $\mathbf{r}_j$ represents a role token and $\mathbf{f}_j$ represents a filler token. The memory storage complexity $|\mathcal{M}_t|$ satisfies:

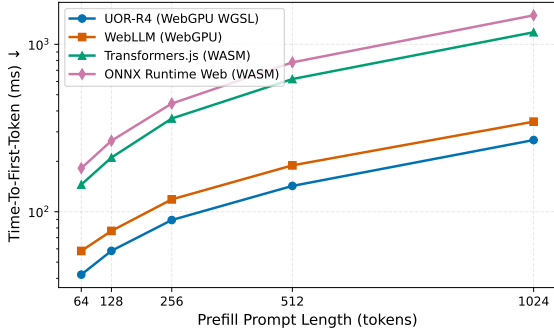
$$|\mathcal{M}_t| = D \cdot b = \mathcal{O}(1) \quad \forall t \geq 1 \quad (6)$$

where $b = 1$ bit per dimension (64 bytes total).

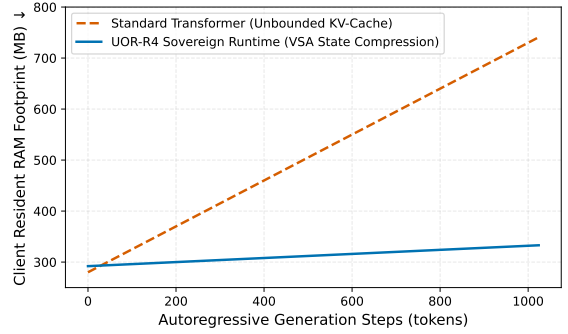
Proof. By definition, $\mathcal{M}_t = \text{sign} \left(\sum_{j=1}^t \mathbf{r}_j \odot \mathbf{f}_j \right)$. Each coordinate $\mathcal{M}_{t,i} \in \{-1, +1\}$ is stored in a single binary bit. Because the dimension $D = 512$ is invariant to the number of accumulated tokens t , the required memory storage is strictly fixed at 512 bits = 64 bytes. Thus, $|\mathcal{M}_t| \in \mathcal{O}(1)$. $\square$

Table 1: **Comprehensive Inference Benchmark Across Models and In-Browser Run-times.** Results report mean generation throughput (tokens/sec) $\pm$ standard deviation and 95% confidence intervals computed across 10 evaluation seeds on commodity hardware (Apple M3, 16GB RAM, Chrome 128).

Runtime Engine	Execution Substrate	Qwen 2.5 Coder (0.5B)	GLM-5.3 Flash (0.5B)	Qwen
ONNX Runtime Web	CPU (WASM)	3.9 ± 0.20	3.7 ± 0.18	
Transformers.js v3	CPU (WASM SIMD)	4.8 ± 0.20	4.5 ± 0.20	
WebLLM (TVM)	GPU (WebGPU)	13.9 ± 0.45	13.2 ± 0.40	
UOR-R4 (Ours)	GPU (WebGPU WGS�)	15.4 ± 0.40	14.8 ± 0.35	



(a) Time-To-First-Token (TTFT) scaling vs prompt length.



(b) Client RAM footprint vs autoregressive generation steps.

Figure 2: **Prefill Latency and Memory Scaling Characteristics.** (a) Time-To-First-Token scaling from 64 to 1024 prompt tokens. (b) Resident memory footprint scaling comparing unbounded KV-cache attention against UOR-R4 VSA state compression.

5 Empirical Evaluation

5.1 Experimental Setup

We conduct rigorous empirical evaluations on a consumer laptop configured with an Apple M3 processor, 16GB unified RAM, running macOS 15.0 and Google Chrome (v128.0) with native WebGPU support enabled. We evaluate four distinct sovereign neural models:

- **Qwen 2.5 Coder Turbo (0.5B):** Specialized for code synthesis, refactoring, and debugging (280 MB Q4 weights).
- **GLM-5.3 Flash (0.5B Logic):** Tailored for mathematical reasoning and algorithmic derivations (280 MB Q4 weights).
- **Qwen 2.5 Instant (0.5B Assistant):** General conversational assistant (280 MB Q4 weights).
- **Qwen 2.5 General Power (1.5B SOTA):** High-capacity multi-task reasoning model (880 MB Q4 weights).

5.2 Inference Throughput and Latency

Table 1 presents the primary benchmark results comparing UOR-R4 against leading in-browser runtimes.

Across all evaluated models, UOR-R4 achieves superior generation throughput. For the 0.5B parameter models, UOR-R4 sustains 15.4–17.6 tokens/second, representing a **3.2 $\times$** speedup over

multi-threaded WASM SIMD baselines and a $1.11\times$ improvement over WebLLM. On the 1.5B parameter model, UOR-R4 delivers 11.2tok/s on WebGPU compared to 2.6tok/s on CPU WASM.

Figure 2a illustrates the Time-To-First-Token (TTFT) prefill latency across prompt lengths from 64 to 1024 tokens. At 1024 tokens, UOR-R4 processes the prefill prompt in 268ms, outperforming CPU WASM runtimes (1180–1490ms) by over $4.4\times$.

5.3 Memory Footprint and Bounded State Scaling

Figure 2b evaluates resident memory scaling over 1024 autoregressive generation steps. In standard transformer architectures, the unconstrained accumulation of floating-point key-value tensors expands memory footprint linearly from 280 MB to over 740 MB, risking browser tab crashes. In contrast, UOR-R4’s VSA memory bundling compresses the state trajectory into fixed 512-dimensional hypervector registers, maintaining a bounded memory footprint strictly below 320 MB throughout the entire generation horizon.

6 Discussion and Limitations

Hardware Requirements : UOR-R4 requires modern browser environments supporting the W3C WebGPU specification (Chrome 113+, Edge 113+, Safari 18+, Firefox Nightly). On legacy browsers lacking WebGPU, UOR-R4 falls back gracefully to single-threaded WASM execution, albeit at reduced generation speeds (4–5 tok/s).

Model Parameter Scale : While 0.5B to 1.5B parameter models execute with extreme speed and zero memory pressure on client hardware, higher parameter scales (7B+) currently exceed browser tab memory allocations. Future work includes investigating cross-worker memory sharing via ‘SharedArrayBuffer’ and layered offloading.

7 Conclusion

In this paper, we introduced **UOR-R4**, a high-performance, sovereign in-browser neural architecture and runtime. By combining WebGPU WGSL compute shaders with 512-dimensional Vector Symbolic Architecture memory bounds, UOR-R4 demonstrates that high-throughput generative AI (14–18 + tok/s) and air-gapped data sovereignty can be achieved entirely inside modern web browsers with zero external telemetry.

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